

Most Asked Java Backend Interview Questions

Real interview-style questions for Java, Spring Boot, Microservices, and Production Scenarios

Core Java

- Difference between HashMap and ConcurrentHashMap?
- How does ConcurrentHashMap work internally?
- Difference between fail-fast and fail-safe iterator?
- Difference between ArrayList and LinkedList?
- Difference between == and equals()?
- Why should hashCode() and equals() be overridden together?
- What is immutability and why is String immutable?
- Difference between checked and unchecked exceptions?
- How JVM memory works?
- Difference between heap and stack memory?
- What causes memory leaks in Java?
- What is garbage collection and how does it work?
- Difference between Runnable and Callable?
- What is synchronization in Java?
- What is volatile keyword?
- What is deadlock and how do you avoid it?
- Difference between sleep() and wait()?
- What is Java Stream API?
- Difference between map() and flatMap()?
- What is CompletableFuture?
- Difference between parallel stream and normal stream?

Spring Boot

- What happens internally when Spring Boot application starts?
- What is dependency injection?
- Difference between @Component, @Service, and @Repository?
- Difference between @RestController and @Controller?
- What is bean lifecycle in Spring?
- What is Spring Boot auto-configuration?
- How does @Transactional work internally?
- What are propagation levels in transactions?
- How do you handle global exception handling in Spring Boot?
- How do you secure REST APIs?
- What is JWT and how does it work?
- Difference between RestTemplate and WebClient?
- What is Spring Boot Actuator?
- How do you monitor Spring Boot applications in production?
- How do you externalize configurations in Spring Boot?
- How do you optimize Spring Boot application startup time?
- How do you implement caching in Spring Boot?
- Difference between @PathVariable and @RequestParam?

Microservices

- What are microservices and why are they used?
- Difference between monolith and microservices?
- What is API Gateway and why is it required?
- How does service discovery work?
- What is Eureka Server?
- How do microservices communicate with each other?
- Difference between synchronous and asynchronous communication?

- What is circuit breaker pattern?
- What is distributed tracing?
- How do you handle distributed transactions?
- What is Saga pattern?
- What is idempotency in microservices?
- How do you prevent cascading failures?
- What is Spring Cloud Config?
- How do you secure communication between microservices?
- How do you deploy microservices using Docker?
- What is Kubernetes and why is it used?
- How do you scale microservices?
- How do you handle versioning in APIs?

Production Issues

- Your application suddenly becomes slow during peak traffic. How will you debug it?
- CPU usage suddenly increased in production. What will you check first?
- Memory usage keeps increasing after deployment. How will you identify the issue?
- One API is taking 15 seconds to respond in production. How will you debug it?
- How do you analyze thread dumps?
- How do you identify slow SQL queries?
- How do you troubleshoot database connection pool exhaustion?
- How do you debug OutOfMemoryError in production?
- Your microservice is failing intermittently. What will you check?
- How do you handle high traffic in a Spring Boot application?
- How do you debug timeout issues between microservices?
- What monitoring tools have you used in production?
- How do you reduce API latency in production?
- How do you handle sudden spike in traffic?

- How do you debug deadlocks in production?
- How do you troubleshoot pod crashes in Kubernetes?
- What logs and metrics do you check first during incidents?

Curated from recent Java backend interview trends and real production scenario discussions.